

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Computer Vision [ECE_4410]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Given a digital video of 10 seconds with frame rate of 60 frames per second, each frame being 512 x 512 x 3 color image, each pixel is represented by 8 bits of resolution. Compute the time required to transmit the video over a channel with transmission rate of 2.097 Mbps, assuming no data compression and ideal conditions. (5)
- A)
- Course Outcome: "CO1" / Cognitive Level: "Understand"
- B) Explain in brief on aspect ratio with respect to any display device. Bring out the rational why any display device does not have same dimension for height and width? (5)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Explain Laplacian of a digital image. Derive the elements of a 3 x 3 Laplacian mask using basic principles. (5)
- A)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Describe the concept of weighted average filter applied in digital image processing with the necessary equations and interpretation. (5)
- Course Outcome: "CO2" / Cognitive Level: "Understand"
- 3) Illustrate the concept of image segmentation. Explain in detail on any two, 3 x 3 masks that can be used in segmenting digital images? (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Demonstrate on how gradient operators can be used in image segmentation. Deduce the mask for gradient operation using fundamental principles? (5)
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Illustrate the concept of class conditional probability density function. Explain how univariate Gaussian data can be modeled using class conditional probability density function. (5)
- A)
- Course Outcome: "CO4" / Cognitive Level: "Analysis"
- B) Analyse the concept of class histogram used in Bayesian learning. (5)
- Course Outcome: "CO4" / Cognitive Level: "Application"

5) Develop the multilayer perceptron architecture with computational steps? (5)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Explain the 2 category SVM model used in pattern classification, with necessary computational expressions. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4410:Computer Vision

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4410:Computer Vision

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	5	25(50%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)